AI Portfolio

AI Architecture Advisor

8-Agent Pipeline — Discovery to CFO Narrative

in 60 seconds, not 5-7 days

Demo Mode | **Claude API** | **OpenAI API**

danvzla.github.io/ai-architecture-advisor

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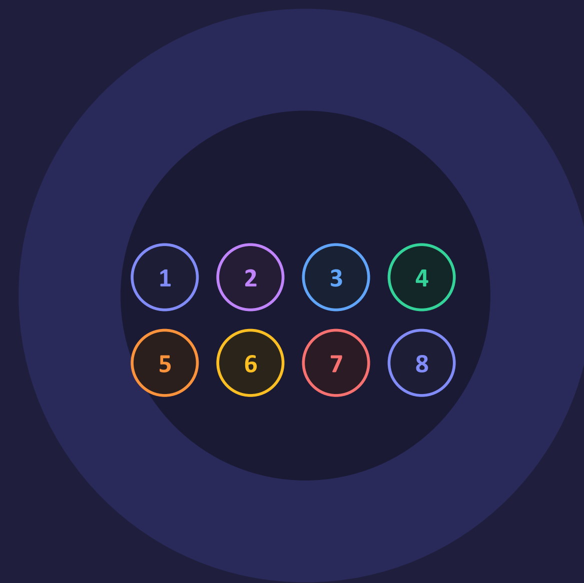

THE PROBLEM

Every enterprise AI engagement starts the same way and takes 5-7 days before any work begins

Day 1-2	Discovery & problem framing	*Senior Architect*
Day 2-3	Reference architecture design	*Principal Architect*
Day 3-4	Architecture Decision Records	*Principal Architect*
Day 4-5	Risk register & governance controls	*Risk Architect*
Day 5-6	30/60/90 day delivery roadmap	*Senior TPM*
Day 6-7	ROI model & CFO executive narrative	*Value Engineer*

5-7 days. Each engagement starts from scratch. ADRs are almost never written until an audit forces it. The ROI case gets assembled after the architecture decision — advocacy, not analysis.

7 DELIVERABLES

Seven enterprise-grade deliverables from one discovery session

1 **Reference Architecture**

Layered target-state diagram with specific technologies per layer

2 **Architecture Decision Records**

Context · decision · rationale · tradeoffs · alternatives rejected

3 **Risk Register & Governance**

Risks ranked by severity, category, likelihood, mitigation, owner

4 **30/60/90 Day Roadmap**

Phased milestones, deliverables, dependencies, go-live criteria

5 **TCO / ROI / NPV Model**

Full financial model with sensitivity analysis at 65/100/125%

6 **CFO Executive Narrative**

Investment justification, urgency driver, recommended decision

7 **AI Readiness Scorecard**

6-dimension maturity score with gap analysis and legend

KEY DESIGN DECISION

The financial math is never AI-generated — only AI-narrated

Step 7 wizard inputs (volume, labor cost, error rate...)

financial.js
Pure deterministic function

Agent 08 receives the calculated numbers and writes the CFO narrative around them

Why this matters:

- Most AI tools let the LLM calculate ROI directly — which means it can hallucinate numbers
- This tool calculates TCO, ROI, NPV, payback, and cost of inaction with deterministic math first
- Agent 08 never does arithmetic — it only writes the narrative around numbers that are already correct
- The CFO narrative is persuasive writing on top of verified math, not a number the AI invented

THREE MODES

Zero setup to live AI generation — one click apart

`No API key required`

Demo Mode

Select Demo → choose a scenario → click Quick Demo. Full blueprint loads in ~8 seconds using a pre-built Telco NOC scenario. All 8 agents animate, all 7 tabs populate.

Best for: Always ready — works offline, works from the desktop

`claude-haiku-4-5`

Claude API

Switch to Claude → enter sk-ant-... key → run. Each of the 8 agents makes an independent API call, results chain forward. Full reasoning grounded in your discovery inputs.

`gpt-4o-mini`

OpenAI API

Switch to OpenAI → enter sk-... key → run. Identical 8-agent pipeline, same 7 output tabs. Provider is swapped by one line in api-client.js.

7-STEP DISCOVERY WIZARD

Mirrors how a senior architect actually runs discovery

1	**Profile**	Industry · function · company size · AI maturity (1-6) · urgency
2	**Problems**	Problems to solve · pain level · stakeholders · description
3	**Outcomes**	Target outcomes · success metrics · impact timeframe · adoption target
4	**Capabilities**	Required AI capabilities · autonomy level · UX · multi-agent need
5	**Data**	Knowledge sources · data quality · deployment model · integrations
6	**Governance**	Industry risk · governance maturity · auditability · internal capability
7	**Value**	Annual volume · effort · labor cost · error rate · implementation cost

⚡ **Step 7 inputs drive the financial model — labor savings, rework reduction, delay avoidance, TCO, ROI, NPV, payback, cost of inaction**

ARCHITECTURE · HLD

Three layers · browser-native · build step for deployment

PRESENTATION

- **index.html — mode selector, scenario dropdown, wizard shell**
- assets/css/styles.css — full design system, dark theme
- Wizard overlay · agent pipeline animation · 7-tab results
- Browser-native · GitHub Pages · zero infra cost

Built via node build.js
dist/index.html self-contained

ORCHESTRATION

- **assets/js/app.js — mode mgmt, pipeline loop, export**
- AGENT_REGISTRY[] — ordered agent array (10 lines)
- Prior results accumulated, passed to each agent
- FinancialModel.calculate() runs before Agent 08

Pure JavaScript
No bundler dependencies

AGENT + API

- **agent-01 → agent-08 (8 individual files)**
- api-client.js — Claude ↔ OpenAI provider swap
- demo-blueprint.js — pre-built output, no API key
- scoring.js + financial.js — pure math functions

1-line provider swap
in app.js

ARCHITECTURE · LLD

One file per agent · 5-attempt JSON repair chain

Agent file interface (identical for all 8)

```
window.Agent04Architecture = {
  id: 'architecture',
  name: 'Architecture Agent',
  mission: 'Target architecture · ADRs',
  tokens: 4500,

  buildPrompt(answers, prior) {
    // wizard answers + prior[discovery]
    // + prior[capability]
  },

  async run(apiKey, answers, prior) {
    const raw = await ApiClient.call(
      apiKey, this.buildPrompt(answers,prior),
      this.tokens);
    return ApiClient.parseJSON(raw);
  },
};
```

JSON Repair Chain

5 attempts — one more than the other AI Portfolio tools (handles whitespace-run collapsing too):

1. **Direct JSON.parse()**
2. **Strip control chars**
3. **Collapse literal newlines**
4. **Collapse whitespace runs**
5. **Close unclosed brackets**

Maintenance rule: adding agent 9 = one new file + one line in agents.js

15 PRE-BUILT SCENARIOS

Broadest vertical coverage across the AI Portfolio

01
Telco Network AIOps Platform
Telecommunications

02
Financial Services GenAI Assistant
Financial Services

03
Healthcare Clinical Decision Support
Healthcare

04
Government Document Intelligence
Public Sector

05
Enterprise IT Operations (ServiceNow)
Technology

06
Retail Customer Experience AI
Retail

07
Manufacturing Predictive Maintenance
Manufacturing

08
Cybersecurity Threat Intelligence
Technology

09
Insurance Claims Processing AI
Financial Services

10
Energy Grid Optimization AI
Energy

11
Legal Contract Intelligence
Professional Services

12
Supply Chain Intelligence
Manufacturing

13
HR Talent Intelligence Platform
Technology

14
Pharma Clinical Trials AI
Healthcare

15
Private Cloud AI Infrastructure
Technology

SAMPLE OUTPUT

Telco Network AIOps Platform — Scenario 01

Financial Model

Annual gross benefit	$845K
3-year ROI	87%
NPV @ 8% discount	$1.1M
Payback period	14 months
Cost of inaction (3yr)	$2.6M

Architecture Pattern Selected

RAG + Agentic AI
with Human-in-the-Loop
and Zero Trust Enterprise Integration

Sample ADR-002

Decision: Primary: Azure OpenAI gpt-4o. Fallback: Anthropic Claude claude-haiku-4-5 via Azure Marketplace.

Rationale: Azure OpenAI meets data residency compliance. Dual-provider fallback ensures 99.9% AI availability.

Alternatives rejected: OpenAI direct API (data leaves Azure boundary). Single provider (unacceptable availability risk for 24/7 NOC).

KEY BENEFITS

What this demonstrates — and what it replaces

~8s

Demo Mode

Full 8-agent blueprint, no API key

8

Chained agents

Each reads all prior agents' output

7

Deliverables

Architecture to CFO narrative

15

Scenarios

Widest vertical coverage in portfolio

5

JSON repair attempts

Most resilient parsing in the portfolio

0

AI-calculated finance

ROI/NPV computed deterministically

LIVE DEMO · NO API KEY REQUIRED

Try it right now

Live demo → danvzla.github.io/ai-architecture-advisor

GitHub → github.com/danvzla/ai-architecture-advisor

- Select Demo Mode → Quick Demo → full 8-agent blueprint in ~8 seconds, no key
- 7 output tabs: Architecture · ADRs · Risk · Roadmap · ROI · CFO Narrative · Scorecard
- 15 pre-built scenarios spanning Telco, Financial Services, Healthcare, Gov't & more
- Financial math computed deterministically — never hallucinated by the AI

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